# Measured cost of cosine-filter ground-state preparation from a matrix-product state: gap, overlap and target in the $J_1$–$J_2$ chain, with a comparison to QETU

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A common route to a ground state on a quantum computer compiles a classical trial state and removes its residual excited weight with a quantum filter. We measure the cost of this route by exact simulation of the Trotterized circuits on the spin-$\frac{1}{2}$ $J_1$–$J_2$ model, with a matrix-product-state (MPS) trial and the one-ancilla cosine filter of Stetcu, Baroni and Carlson (SBC). A controlled study at $N = 8$–$14$ holds the gap (0.30–0.55) and the trial overlap fixed separately and shows that they act on different resources: the gap sets the length of the filter and the depth of the circuit, while the overlap enters the circuit only logarithmically and the cost per success as $1/\gamma$. The long, smooth SBC pulses tolerate Trotter steps of order $1/J_1$, so the measured cost per success grows only as $N^{1.50}\Delta^{-1.15}$. Against a Chebyshev (QETU) filter under identical accounting, SBC is cheaper at $\epsilon \approx 10^{-2}$, the two tie at $10^{-3}$, and QETU is cheaper from $10^{-4}$. Against compiling a better MPS, filtering wins on chains only as a last-mile polisher (18 of 63 targets), but at 27 of 28 targets on $4 \times 3$ and $4 \times 4$ cylinders, where a snake MPS needs a fast-growing bond dimension. Planning from DMRG alone, with the Trotter step set by energy convergence, met the target in all 182 cases checked against exact diagonalization, and DMRG error bars from the bond-dimension ladder covered the exact energies in all 75 tests; a 10–25% gap overestimate is the error that breaks a filter.

## I. INTRODUCTION

A common route to a ground state on a quantum computer starts classically: an approximate method produces a trial state that can be compiled into a short circuit, and a quantum subroutine removes its residual excited-state weight. For one-dimensional systems the natural classical engine is DMRG [2, 3], which yields a matrix-product state (MPS) whose low-bond-dimension truncations compile exactly into staircase circuits [4]. The quantum step can be phase estimation [5], a polynomial filter implemented by quantum singular-value transformation [6–8], or a product of cosine filters implemented with a single ancilla [1, 12]. The last option, the projection algorithm of Stetcu, Baroni and Carlson (SBC) [1], needs little beyond time evolution: each pulse is one controlled time evolution between ancilla rotations and a measurement.

Resource estimates for such filters usually rest on worst-case bounds, which are loose by orders of magnitude for Trotter errors [27]. Here we measure instead. For the open $J_1$–$J_2$ Heisenberg chain and small cylinders we simulate every circuit exactly and find, for each case, the shortest filter and the coarsest Trotter step that reach a target infidelity $\epsilon$. We ask three questions. *What sets the cost?* The spectral gap $\Delta$ and the trial overlap $\gamma$ normally change together with system size $N$; we decouple them, holding the gap fixed by tuning $J_2$ and the overlap fixed by a one-parameter rotation of the trial circuit. *Which filter is cheaper?* We compare SBC with a Chebyshev polynomial filter (QETU) [8] under identical accounting. *When does filtering pay?* We compare filtering a cheap MPS with compiling a better one. Finally, we check that the filter can be planned from DMRG alone, by running the whole workflow on DMRG inputs, with unconverged DMRG and with deliberately wrong inputs, on systems where exact diagonalization verifies the result.

*Relation to prior work.* The filter is SBC's [1]; it shares its circuit with the rodeo algorithm [12, 18], which draws the times at random. Our "dyadic" schedule, a halving ladder of times, is the one SBC and Rule *et al.* use for quantum-number projection [1, 19], and for energy it is a finite rodeo super-iteration [20, 21]. That filter depth scales as $1/\Delta$ and repetitions as $1/\gamma$ is the known complexity of filter methods [5, 6]. QETU has been costed with Trotterized evolution [22] and run on small Hubbard models [23, 24]. Resource studies that compile a classical trial and refine it exist for quantum chemistry [10, 11, 25], and Watanabe *et al.* [26] finish an early-stopped MPS encoder with probabilistic imaginary-time evolution on Heisenberg chains. What is new here is (i) a controlled separation of gap and overlap at fixed size, which turns the known scaling into measured constants and exponents, (ii) a measured factorization of the cost into filter length, Trotter step and gates per step, (iii) a like-for-like measured comparison of cosine and Chebyshev filters, including the Chebyshev filter's query length, (iv) a measured benchmark of filtering a cheap MPS against compiling a better one, and (v) DMRG-only runs and sensitivity studies verified against exact diagonalization. The code and data are at `https://github.com/aaronseymour7/FilterPipeline`.

## II. FILTERS, CIRCUITS AND HOW WE MEASURE THEM

*Spectral window.* With exact energies $E_0$, $E_1$ and an upper bound $E_{\max}$ on the spectrum, the scaled Hamiltonian $H_s = (H - E_0)/W$, $W = E_{\max} - E_0$, has its ground energy at 0 and every other level in $[\Delta, 1]$, $\Delta = (E_1 - E_0)/W$. To leave a fraction $\epsilon$ of excited weight in a trial state with ground-state overlap $\gamma$, a filter $F$ must suppress the excited levels relative to the ground

state by $\eta$ with $\eta^2 \approx \gamma\epsilon/(1-\gamma)$.

*SBC filter.* One pulse [1] applies $\mathsf{H}_a\, e^{-itH_s\otimes Z_a}\, R_z^{(a)}(2\phi)\, \mathsf{H}_a$ to the system and an ancilla in $|0\rangle$; its ancilla-$|0\rangle$ block is $\cos(H_s t+\phi)$. Keeping outcome 0 after $m$ pulses applies

$$F(H_s)=\prod_{i=1}^{m}\cos(H_s t_i+\phi_i), \qquad T=\sum_i |t_i|, \quad (1)$$

with success probability $p \approx \gamma F(0)^2$. We use two schedules. *Dyadic*: $r$ repetitions of the halving ladder $t=\tau,2\tau,\ldots,2^{K-1}\tau$, $\tau=\pi/(1+\Delta)$, $\phi=0$. One block is a Dirichlet kernel,

$$\prod_{k=0}^{K-1}\cos(2^k E\tau)=\frac{\sin(2^K E\tau)}{2^K\sin(E\tau)}, \quad (2)$$

equal to 1 at $E=0$ and at most $1/(2^K\sin\theta)$, $\theta=\pi\Delta/(1+\Delta)$, on $[\Delta,1]$, so $T\approx(e/\Delta)\ln(1/\eta)$, the optimal scaling [6, 20], with no optimization and $F(0)=1$. *Optimized*: SBC's energy recipe [1], times and phases that maximize $F(0)^2$ at fixed $(T,m)$ subject to suppression $\eta$ on the whole of $[\Delta,1]$; nonzero phases trade success probability for a shorter filter.

*Trotterization.* Each controlled evolution $e^{-it_iH_s\otimes Z_a}$ is approximated by $k_i=\lceil|t_i|/\delta t\rceil$ second-order (Strang) steps with a common step $\delta t$; the identity term becomes one exact ancilla $R_z$ per pulse.

*Gate counts.* We assume all-to-all connectivity and Clifford+$R_z$ gates. For a Heisenberg bond $(i,j)$ the three commuting controlled terms form one exact block,

$$e^{-ict(X_iX_j+Y_iY_j+Z_iZ_j)\otimes Z_a}=V^\dagger e^{-ict(Z_iZ_a+Z_jZ_a-Z_iZ_jZ_a)}V,$$
$$V=\mathsf{H}_i\,\mathrm{CNOT}_{ij}, \quad (3)$$

whose diagonal factor is a phase polynomial on three parities [29]: 6 CNOTs and 3 $R_z$, against 12 CNOTs for three separate Pauli rotations. Control by an anticommuting Pauli string [8, 9] is not available, because no Pauli product anticommutes with all of $XX$, $YY$ and $ZZ$ on a bond. $k$ Strang steps cost $k(2B-2)+1$ blocks for $B$ bonds, and each pulse adds one ancilla $R_z$. An MPS of bond dimension $2^b$ is a staircase of $N-b$ unitaries on $b+1$ qubits, each costed by the quantum Shannon decomposition bound [13]. The cost of one attempt is the trial circuit plus the filter, and the expected cost per success is that divided by $p$ (repeat-until-success).

*Measured cost.* For each case the exact filter is applied to the trial state to find the shortest filter that reaches infidelity $\epsilon$ (dyadic: every ladder $(K,r)$; optimized: SBC's design with the true gap and overlap). At that filter, the coarsest common Strang step at which the full Trotterized circuit, simulated on the state vector, still reaches $\epsilon$ is found by bisection, and the cheaper of the two shortest dyadic ladders is kept. Optimized designs that barely pass the ground state ($p<\gamma/2$) are repeated over optimizer seeds (21 of 286 points). Every measured cost therefore uses the exact ground state: it is what the circuit has to do, not a rule an experiment could apply unchanged; Sec. VII plans from DMRG alone.

## III. MODEL

We study the open spin-$\frac{1}{2}$ chain

$$H=J_1\sum_{i=1}^{N-1}\mathbf{S}_i\cdot\mathbf{S}_{i+1}+J_2\sum_{i=1}^{N-2}\mathbf{S}_i\cdot\mathbf{S}_{i+2}, \quad (4)$$

$J_1=1$, which is gapless at $J_2=0$ and dimerized above $J_2\approx 0.2411$ [14, 15]; it has $B=2N-3$ bonds for $J_2\neq 0$. Spectra are computed exactly per $S^z$ block. MPS trials are the variationally optimal MPS at fixed bond dimension, computed with quimb [16].

## IV. GAP AND OVERLAP AS INDEPENDENT INPUTS

Along any natural family of chains the gap and the overlap move together: at fixed $J_2$ the gap closes with $N$, and at fixed bond dimension the overlap decays with $N$. We therefore construct a family in which they are set independently at every $N$.

*Construction.* For each $N\in\{8,10,12,14\}$ and target gap $g=E_1-E_0\in\{0.30,0.35,\ldots,0.55\}$, $J_2$ is found by bisection on exact diagonalization so that the singlet–triplet gap equals $g$ to $10^{-9}$; pairs without a solution below the dimerized regime are left out. For all tuned couplings the lowest $S^z=0$ excitation coincides with the lowest triplet to $10^{-7}$, so $g$ is the true gap. The trial is the variational $\chi=16$ MPS, prepared by its staircase circuit, followed by $R_y(\theta)$ on every even site; rotating one sublattice breaks the singlet symmetry and lowers $\gamma$ smoothly, and $\theta$ is set by bisection to hit the target $\gamma$. The trial circuit is therefore the same along an overlap scan. Study A fixes $g\in\{0.40,0.50\}$ and scans $\gamma\in\{0.03,0.1,0.3,0.5,0.7,0.9\}$; study B fixes $\gamma\in\{0.1,0.5\}$ and scans $g$; both at $\epsilon\in\{10^{-2},10^{-3},10^{-4}\}$, 222 cases in all, each with the dyadic and optimized schedules and with QETU (Sec. V). Costs are for the filter alone unless stated.

### A. The gap sets the circuit, the overlap sets the repetitions

Figures 1 and 2 show CNOTs, $R_z$ rotations and evolution time per shot, and CNOTs per success, along both scans.

*Per shot, the overlap enters only logarithmically.* At fixed gap, lowering $\gamma$ thirtyfold (0.9 to 0.03) raises the CNOTs per shot of the dyadic filter by only 1.2–3.0× and its evolution time by 1.5–3.0×. The filter must reach $\eta^2=\gamma\epsilon/(1-\gamma)$, and a worse trial adds only $\ln 290\approx 5.7$ to $L=\ln[(1-\gamma)/(\gamma\epsilon)]$, already 2–13 here. The dyadic filter grows in whole pulses, so its per-shot cost moves in steps.

*Per shot, the gap enters as a power.* At fixed overlap, closing the gap by 1.8× (0.55 to 0.30) raises the CNOTs

per shot by 2.7–4.3× and the evolution time by 1.5–2.8×, about as $1/\Delta$. Both behaviors are the textbook $T = O(\Delta^{-1} \ln \eta^{-1})$ [5, 6]; the controlled study shows that real Trotterized circuits follow it at fixed $N$ with the inputs varied separately.

*Per success, the overlap returns as* $1/\gamma$. Every run succeeds with probability $p \approx \gamma F(0)^2$, with $F(0) = 1$ for the dyadic filter, so the cost per success carries $1/\gamma$. The gap decides the depth of the coherent circuit, and with it whether the filter fits a coherence or error budget at all; the overlap decides how often that circuit has to be run.

*$R_z$ counts follow CNOT counts.* Each bond block has 3 $R_z$ and 6 CNOTs and each pulse adds one rotation, so $R_z/\mathrm{CNOT} = 0.50$ throughout, and every statement about CNOTs holds for the arbitrary rotations, and so for the $T$ count of a fault-tolerant implementation up to the synthesis factor.

### B. Mechanism: filter length, Trotter step and gates per step

The per-shot cost factorizes as

$$\text{CNOTs per shot} = \frac{T}{\delta t} \times c_{\text{step}}(N), \tag{5}$$

and each factor answers to different inputs (Fig. 3).

*Filter length.* The physical evolution time depends on the gap and $L$ only: a fit gives $T \propto \Delta^{-1.19} L^{0.77} N^{-0.24}$ for the dyadic filter, so at fixed absolute gap the filter length does not depend on the size of the system.

*Trotter step.* The filter tolerates very coarse steps: median $\delta t = 1.25,\ 0.56,\ 0.30$ (in $1/J_1$) at $\epsilon = 10^{-2},\ 10^{-3},\ 10^{-4}$, larger at larger gap. Its pulses are long and smooth, and the filtered state lives at the bottom of the spectrum, where Trotter errors are much smaller than their worst case [17, 28]. The step shrinks roughly as $\epsilon^{0.3}$ and barely depends on $N$, so the number of steps per shot grows as $L^{0.71}$ and $N^{0.08}$.

*Gates per step.* One Strang step costs a median $20\,N$ CNOTs for the two-neighbor chain ($6(2B-2)$ with $B = 2N-3$ bonds). With a step count nearly independent of $N$, the CNOTs per shot grow as $N^{1.30}$, almost entirely through the gates per step.

### C. Cost laws

Fitting the expected CNOTs per success (filter and trial circuit) to

$$C \approx a\, N^{\alpha}\, \Delta^{-\beta}\, L^{\lambda}\, \gamma^{-g}, \qquad L = \ln \frac{1-\gamma}{\gamma\epsilon}, \tag{6}$$

over all 222 cases gives Table I. For the dyadic filter

$$C \approx 10.1\, N^{1.50} \Delta^{-1.15} L^{1.41} \gamma^{-0.78}, \tag{7}$$

within a typical factor of 1.23. The $\gamma$ exponent is below one because at coarse steps each Trotterized pulse is slightly off-diagonal in the eigenbasis and moves some weight into or out of the ground state ($p/\gamma$ ranges from 0.76 to 1.24), so part of the $\gamma$ dependence is absorbed into $L$. The optimized filter grows more steeply with the gap ($\Delta^{-1.36}$), because its design is matched to the gap, and less steeply with $L$; QETU follows its law most closely, to 1.04×. These exponents summarize $N = 8$–$14$ and $g = 0.30$–$0.55$ and are not asymptotic laws. Appendix A compares the two SBC schedules and maps the cost over the $(\Delta, \gamma)$ plane and over $(N, \epsilon)$.

## V. COMPARISON WITH A CHEBYSHEV FILTER (QETU)

QETU [8] uses $d$ queries of $e^{-i\tau H_s \otimes Z_a}$ interleaved with ancilla rotations and a single post-selection, so that the ancilla-0 block is $p(\cos \tau H_s)$ for a real even polynomial $p$ of degree $d$ with $|p| \le 1$. With the stop band mapped to $[0, x_s]$, $x_s = \cos(\tau\Delta)$, we take the extremal Chebyshev filter

$$p(x) = \frac{T_d(x/x_s)}{T_d(1/x_s)}, \tag{8}$$

which has $p$ equal to 1 at the ground state and needs $T = d\tau \approx \Delta^{-1} \ln(2/\eta)$, two to three times less evolution than the dyadic ladder. Each query is Trotterized like an SBC pulse and costed in the same controlled form (Sec. II); the classical phase-finding step costs no quantum gates and is not counted. By simulation on exact spectra for $N = 8$–$16$ (675 cases), the smallest degree that reaches $\epsilon$ is set by the gap and the target alone, $d_{\min}\Delta \simeq (2/\pi)\cosh^{-1}(1/\eta)$, whatever the shape of the trial's excited weight, because the Chebyshev filter suppresses every level above the gap equally.

*Query length* $\pi/2$. With $\tau = \pi/2$, the choice of Dong, Lin and Tong [8], QETU is cheaper than the dyadic filter in 0 of 74 cases at $\epsilon = 10^{-2}$, 9 of 74 at $10^{-3}$ and 56 of 74 at $10^{-4}$. The reason is the Trotter step, not the filter: QETU needs less evolution time (median 0.65× dyadic), but one Strang step per short query, $\delta t = \tau/W \propto 1/N$, so its step count is a median 1.40× the dyadic one and its CNOTs grow as $N^{2.22}$ against $N^{1.30}$. The long dyadic pulses tolerate steps of order $1/J_1$, but these shrink as roughly $\epsilon^{0.3}$, and by $\epsilon = 10^{-4}$ the shorter filter wins.

*Longest admissible query.* Any $\tau \le \pi/(1+\Delta)$ keeps every excited level inside $|\cos \tau E| \le \cos \tau\Delta$, and one Strang step per query still suffices at longer queries. Redoing every QETU search at $\tau = 3\pi/4$ and $\pi/(1+\Delta)$ (467 cases) lowers its cost to a median 0.68 (range 0.29–4.28) of the $\tau = \pi/2$ cost at $\tau = \pi/(1+\Delta)$. With QETU at its best $\tau$ (Table II) the SBC filters keep $\epsilon = 10^{-2}$ (QETU cheapest in 7 of 74), the two families split $10^{-3}$ (38 of 74), and QETU is the cheapest filter in 65 of 74 at $10^{-4}$. The fair statement is that SBC wins at loose targets, the two are even at $10^{-3}$, and the Chebyshev filter wins from $10^{-4}$ on.

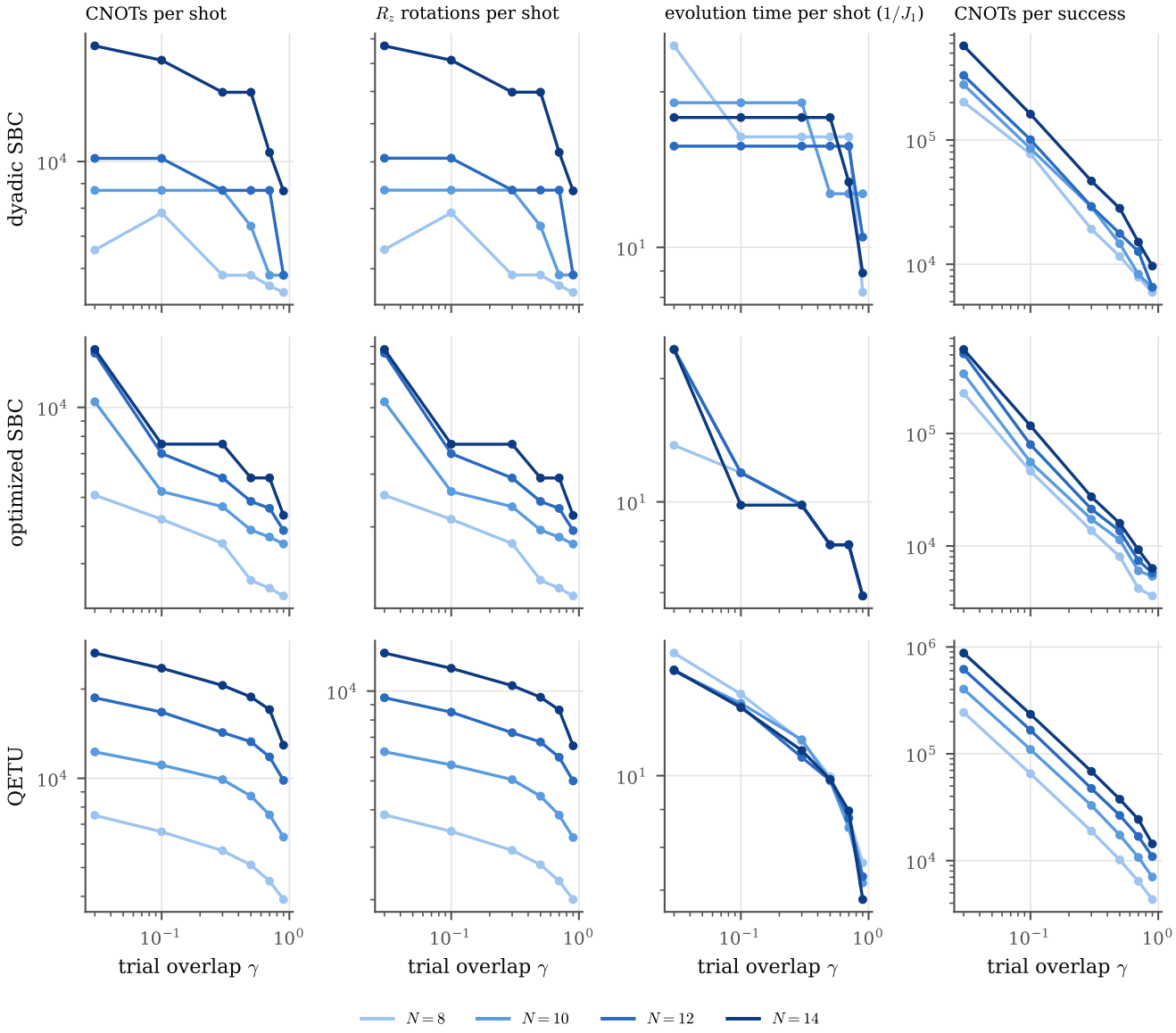



FIG. 1. Study A: filter-only resources at a gap held at $g = 0.40$ for every $N$, $\epsilon = 10^{-3}$, against the trial overlap, measured by exact simulation. Rows: dyadic SBC, optimized SBC, QETU (Sec. V). Columns 1–3 per shot, column 4 per success. Source: `results/gapov.jsonl`, `results/gapov_filter.json`.

## VI. BENCHMARK AGAINST THE MPS ALONE

The alternative to filtering a cheap trial is compiling a better MPS. We compare the two at equal target infidelity: for each system every MPS trial with $\chi = 2, \ldots, 64$ is decomposed on the exact spectrum, and for each $\epsilon \in [10^{-5}, 10^{-2}]$ the cheapest route is found, the smallest MPS that meets $1 - \gamma \leq \epsilon$ alone, against the cheapest smaller MPS followed by a measured SBC or QETU filter, with cost (trial + filter)$/p$. Both routes use the staircase MPS cost of Sec. II.

*Chains.* On nine chains ($N = 12$, 14, 16 at $J_2 = 0$, 0.2411, 0.4; Fig. 4) compiling the MPS is cheaper in most cases: the filtered route wins 18 of 63 targets, by 1.1–3.2×. Every win filters a trial that misses the target by $(1 - \gamma)/\epsilon \leq 8.8$, with a dyadic ladder of one to four pulses (1–21 Strang steps, $p \geq 0.988$), and all 90 candidate routes with a larger ratio lose, by a median 20×. The filter acts as a last-mile polisher: MPS cost jumps about fourfold per doubling of $\chi$, so a target just below what a given $\chi$ reaches forces the MPS-alone route up a bond dimension, while a short filter removes the small residual of the lower one. The wins are therefore tied to the bond-dimension grid rather than to $N$ or $J_2$. The optimized design never wins, because suppression over the whole band above $\Delta$ needs $T \gtrsim \pi/\Delta$ where one dyadic pulse suffices, and QETU beats the MPS alone at 11 tar-

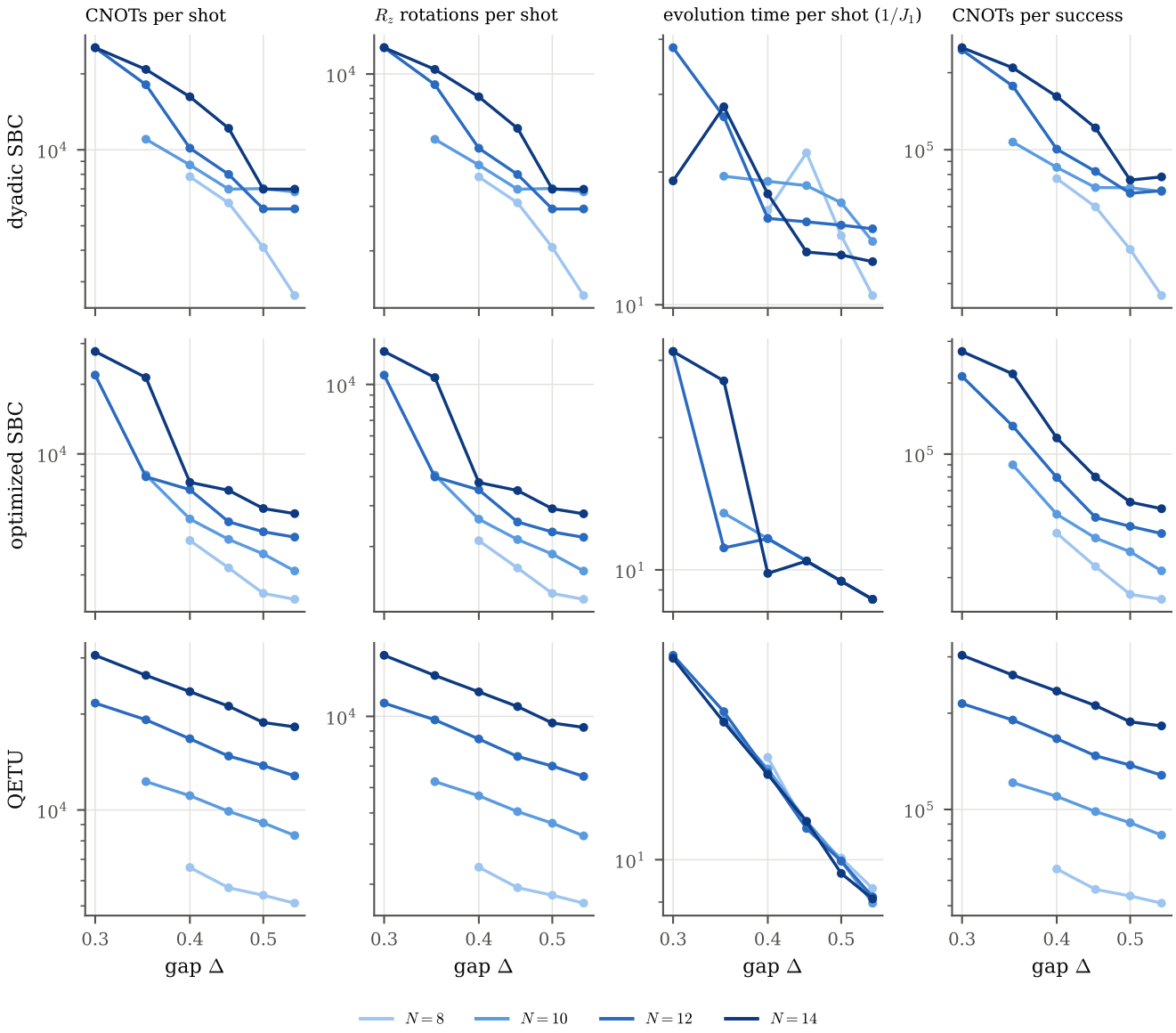



FIG. 2. Study B: as Fig. 1 at an overlap held at $\gamma = 0.10$ for every $N$, against the gap (held fixed across $N$ by tuning $J_2$).

TABLE I. Fits of the measured expected CNOTs per success to Eq. (6) over all 222 gap/overlap cases. "rms"/"max": typical and worst factor between a case and its fit. Source: `results/gapov.jsonl`.

| | cases | $a$ | $\alpha$ | $\beta$ | $\lambda$ | $g$ | rms | max |
|---|---|---|---|---|---|---|---|---|
| dyadic, measured | 222 | $1.0 \times 10^1$ | 1.50 | 1.15 | 1.41 | 0.78 | 1.23 | 1.9 |
| optimized, measured | 222 | 9.8 | 1.50 | 1.36 | 1.24 | 0.83 | 1.30 | 2.3 |
| QETU, measured | 222 | $1.3 \times 10^1$ | 2.09 | 0.74 | 0.68 | 0.99 | 1.04 | 1.2 |

gets but is never the cheapest route.

*Cylinders.* On $4 \times 3$ and $4 \times 4$ cylinders (periodic around the four-site circumference) at $J_2 = 0$ and 0.3, with snake-path MPS trials (Fig. 5), the picture reverses. A snake MPS struggles here: at $4 \times 4$, $J_2 = 0$, $\chi = 16$ misses $\epsilon = 10^{-2}$ ($1 - \gamma = 1.1 \times 10^{-2}$), $\chi = 32$ stalls at $7 \times 10^{-4}$ and no $\chi \leq 64$ reaches $10^{-5}$, while the compile cost grows about fourfold per doubling (5 328, 20 548 and 76 600 CNOTs for $\chi = 16$, 32, 64). The filtered route is cheaper at 27 of the 28 cylinder targets, by 1.1–8.8×,

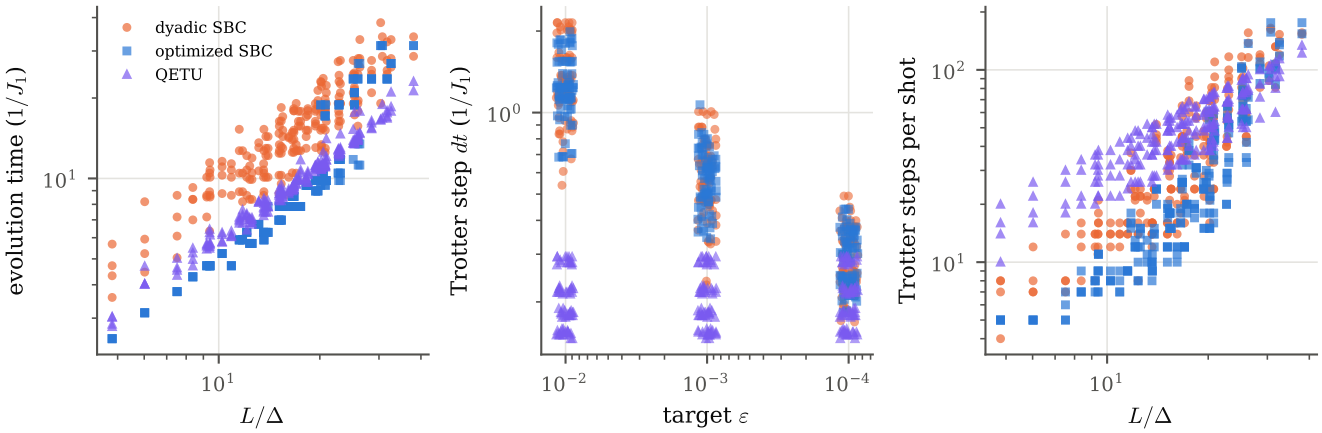



FIG. 3. All 222 cases, measured. (a) Evolution time against $L/\Delta$, $L = \ln[(1-\gamma)/(\gamma\epsilon)]$. (b) Coarsest passing Trotter step against $\epsilon$ (jittered horizontally). (c) Trotter steps per shot against $L/\Delta$. Orange circles: dyadic SBC; blue squares: optimized SBC; purple triangles: QETU.

TABLE II. Cheapest measured filter in the controlled study (74 cases per target; dyadic/optimized/QETU) with QETU at $\tau = \pi/2$ and at its best query length $\tau \in \{\pi/2, 3\pi/4, \pi/(1+\Delta)\}$. Source: `results/qetu_tau.jsonl`.

| target | $\tau = \pi/2$ | best $\tau$ |
|---|---|---|
| $\epsilon = 1 \times 10^{-2}$ | 23/51/0 | 19/48/7 |
| $\epsilon = 1 \times 10^{-3}$ | 16/57/1 | 4/32/38 |
| $\epsilon = 1 \times 10^{-4}$ | 5/32/37 | 0/9/65 |

and is the only MPS-based route at $10^{-5}$ on $4\times4$. Every win uses the dyadic ladder, mostly on a $\chi = 16$ trial, with $p \geq 0.77$ and 1–46 Strang steps, and recovers factors $(1-\gamma)/\epsilon$ up to $10^3$. The filter's depth grows only as $\ln(1/\epsilon)$ at fixed gap, while the MPS needed to reach $\epsilon$ alone grows fourfold per bond-dimension step. All trials are snake MPS, a weak choice in two dimensions, and the staircase cost charges every site a unitary on $b+1$ qubits even where the bond is smaller, which favors the filtered route most where its trial is cheap.

## VII. PLANNING FROM DMRG ALONE, VERIFIED AGAINST EXACT DIAGONALIZATION

The measured costs above pick the filter and the Trotter step with the exact ground state. Here we use only what an experiment would have, DMRG, on systems small enough that exact diagonalization (ED) can check the outcome at the end. The DMRG front end supplies $E_0$, $E_1$ (the lowest $S^z = 1$ state), an MPS trial and its overlap $\gamma$ with the reference MPS, each with an error bar equal to its change between the last two bond dimensions. The filter is designed from these estimates as in Sec. II, and ED enters only to compute the final infidelity.

*End to end.* For every chain ($N = 12$, 14, 16) and cylinder ($4\times3$, $4\times4$) target of Sec. VI, $\epsilon = 10^{-5}$–$10^{-2}$, with the SBC (dyadic or optimized) and QETU filters (182 cases; `run_aposteriori.py`), the filter is designed from the DMRG inputs and the Trotter step is chosen without the ground state: starting from $\delta t = 4/J_1$, halve it until the post-selected energy changes by at most $\epsilon\Delta/2$ between successive steps, and keep the finer one. Because the leaked weight sits at least $\Delta$ above $E_0$, an infidelity $\epsilon$ moves the energy by at least $\epsilon\Delta$, so the rule errs on the safe side. Checked against ED, it met the target in 182 of 182 cases, at a median 1.9 (SBC) and 1.9 (QETU) times the cost with the coarsest step that passes. Its price is the calibration: resolving $\epsilon\Delta/2$ in the three measurement bases of the Heisenberg energy takes a median $2\times10^{10}$ post-selected shots per case, about $10^7$ at $\epsilon = 10^{-2}$ and growing as $\epsilon^{-2}$, which pays off only if the circuit is then reused many times. At these sizes converged DMRG reproduces the ED energies to $10^{-11}$, so this test checks the design and the step rule rather than DMRG errors; the next two tests target those.

*Unconverged DMRG.* To test the error bars themselves we stopped the DMRG ladder at each of $\chi = 8$, 16, 32, 64 and 128 on 15 systems with exact spectra (chains $N = 12$–16, cylinders $4\times3$–$4\times5$, all $J_2$; 75 cases; `run_margin.py`). Without error bars the DMRG energies are consistent with ED in only 29 of 75 cases, and the DMRG gap is up to 17% too large, the dangerous direction, since both DMRG energies are variational upper bounds. With the error bars the intervals contain both exact energies in 75 of 75. Filters designed from each stopped ladder for the four cylinders with full spectra, at $\epsilon = 10^{-2}$–$10^{-4}$ and every schedule, were applied exactly to the trial's spectral weights: all 396 met the target. Once DMRG has converged ($\chi \geq 64$) the error bars cost

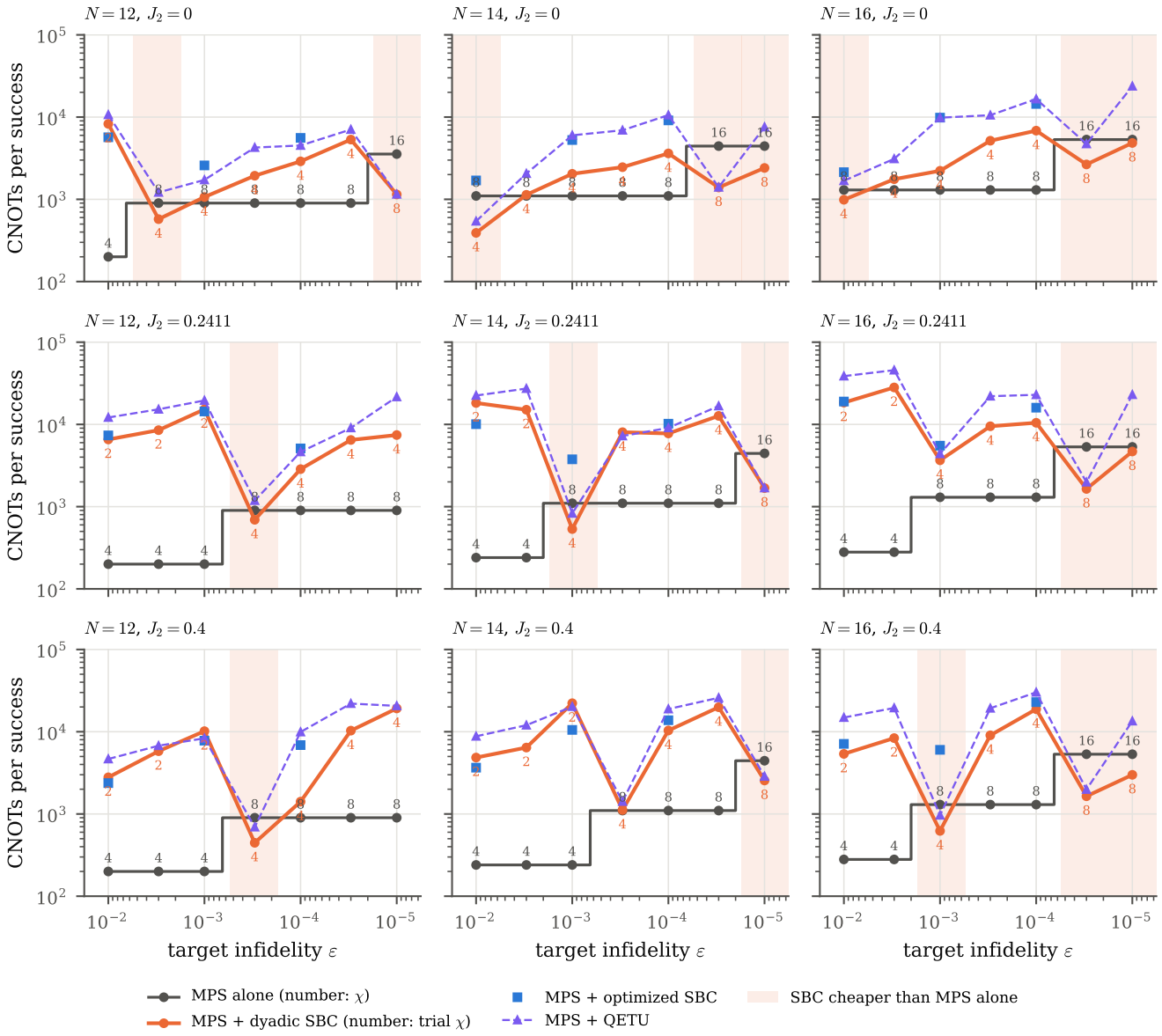



FIG. 4. Measured cheapest route to infidelity $\epsilon$ on chains of $N = 12, 14, 16$ (columns) at $J_2 = 0, 0.2411, 0.4$ (rows), all by exact simulation. Grey: cheapest MPS that meets $1 - \gamma \le \epsilon$ alone (number: $\chi$). Orange: cheapest smaller MPS followed by the dyadic SBC filter (number: trial $\chi$); blue squares: optimized SBC (designed at $\epsilon = 10^{-2}, 10^{-3}, 10^{-4}$ only); purple: QETU. Shaded: the SBC route is cheaper. Cost is (trial + filter CNOTs)/$p$. Sources: `results/emp_actual1d.jsonl` and, for $N = 16$, $J_2 = 0.2411$, `results/emp_actual.json`.

a median factor 1.00 (at most 1.4); at $\chi = 32$ a median 1.6; at $\chi \le 16$ they make the energy window inconsistent in 58 of 72 cases, which the pipeline reports instead of returning a filter.

*Deliberately wrong inputs.* Finally we perturbed the inputs by hand for every $N \le 14$, all three couplings and the dyadic, optimized and QETU filters at $\epsilon = 10^{-2}$, with a conservative Trotter step so that only the filter design is tested, and simulated each circuit on the true trial state (Table III). With exact inputs all 36 filters meet the target. Overestimating the gap moves the stop-band edge above the first excitation: by 10% it breaks the target in 2 of 36 cases (marginally, worst $1.1 \times 10^{-2}$), by 25% in 10 and by 50% in 16. The dyadic filter is the most tolerant (0 misses at 25%), because its suppression is far below the target over most of the band; the optimized and Chebyshev filters, designed to sit right at the target, have no slack. Underestimating the gap is safe and cheap: assuming 0.5 of it costs 2.9×. Overestimating $\gamma$ broke the target in 12 of 72 cases, every one because the inflated overlap crossed $1 - \epsilon$ and the pipeline dropped the filter altogether. The gap is therefore the input to protect:

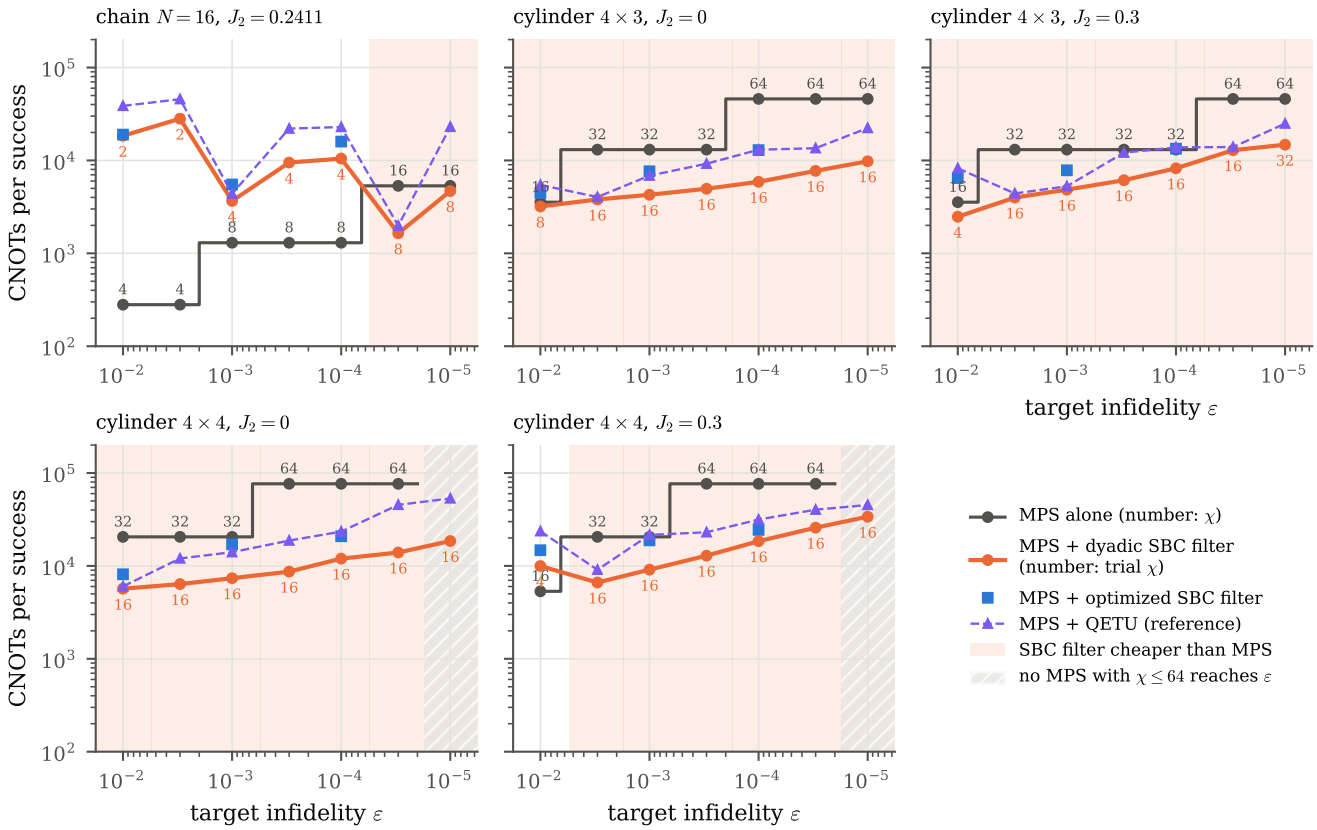



FIG. 5. Measured cheapest route to infidelity $\epsilon$ on the 16-site chain and on small cylinders. Grey: the cheapest MPS that meets the target alone (number: $\chi$). Orange: the cheapest smaller MPS followed by the dyadic SBC filter (number: trial $\chi$); blue squares: optimized SBC; purple: QETU. Shaded: the filtered route is cheaper; hatched: no MPS with $\chi \leq 64$ reaches $\epsilon$. Every cost is (trial + filter CNOTs)/$p$ from exact simulation. Source: `results/emp_actual.json`.

a margin on $E_1$ costs little, and the DMRG error bars above supply one.

## VIII. DISCUSSION

*What sets the cost.* The gap determines the coherent circuit: evolution time $\propto L/\Delta$, independent of $N$ at fixed gap, and CNOTs and $R_z$ per shot that scale as a power of $1/\Delta$. The overlap enters the circuit only through a logarithm and the expected cost through $1/\gamma$, so the filter tolerates a poor trial by paying in repetitions rather than depth; amplitude amplification [30, 31] would turn $1/\gamma$ into $1/\sqrt{\gamma}$ at the price of a deeper coherent circuit, and is not modeled. The target enters through $L$ and, because the tolerated Trotter step shrinks as $\epsilon^{0.3}$, through the step count.

*Which filter.* The long, smooth SBC pulses tolerate Trotter steps of order $1/J_1$, which is why the cosine filter wins at loose targets even though it needs more evolution time than a Chebyshev filter; from $\epsilon = 10^{-4}$ the shorter Chebyshev filter wins. Among the SBC schedules the optimized design is cheaper in most cases (median 0.89× dyadic) but needs a nonconvex design that occasionally fails, while the dyadic ladder needs no design at all.

*When filtering pays.* A filter pays where the trial is hard to improve. For chains the MPS infidelity falls quickly with $\chi$, so the filter helps only as a last-mile polisher; on small cylinders, where a snake MPS needs a fast-growing bond dimension, it wins almost everywhere.

*Measured costs and what an experiment pays.* The benchmarks use the exact ground state to pick the shortest sufficient filter and the coarsest sufficient Trotter step. Planned from DMRG alone, with the step set by energy convergence, the filter met the target in every case we could check, at about twice the measured gate cost, but the calibration costs $\epsilon^{-2}$ shots. The input to protect is the gap: DMRG overestimates it, a 10–25% overestimate breaks the optimized and Chebyshev filters, and the bond-dimension ladder supplies a margin that covered the exact energies in every test at little cost. All of this was verified at sizes where ED is available; at larger sizes the same workflow runs, but its outcome can no longer be checked.

*Limitations.* Noiseless simulation; all-to-all connectivity; repeat-until-success rather than early abort; trial cir-

TABLE III. Gap and overlap overestimation for every $N \in \{8, 10, 12, 14\}$ and $J_2$ (12 instances), $\epsilon = 10^{-2}$, second order: number of plans that miss the target and the worst measured infidelity. $f$: assumed over true gap or $\gamma$. Source: `results/ext_sensitivity.json`.

| | | dyadic | | optimized | | QETU | |
|---|---|---|---|---|---|---|---|
| input | $f$ | miss | worst | miss | worst | miss | worst |
| gap | 1 | 0/12 | $1.7 \times 10^{-3}$ | 0/12 | $4.2 \times 10^{-3}$ | 0/12 | $2.3 \times 10^{-3}$ |
| gap | 1.1 | 0/12 | $6.5 \times 10^{-3}$ | 1/12 | $1.1 \times 10^{-2}$ | 1/12 | $1.0 \times 10^{-2}$ |
| gap | 1.25 | 0/12 | $6.8 \times 10^{-3}$ | 5/12 | $2.6 \times 10^{-2}$ | 5/12 | $3.0 \times 10^{-2}$ |
| gap | 1.5 | 2/12 | $6.3 \times 10^{-2}$ | 7/12 | $6.8 \times 10^{-2}$ | 7/12 | $9.2 \times 10^{-2}$ |
| gap | 2 | 8/12 | $1.4 \times 10^{-1}$ | 7/12 | $1.5 \times 10^{-1}$ | 11/12 | $2.2 \times 10^{-1}$ |
| gap | 3 | 11/12 | $2.2 \times 10^{-1}$ | 12/12 | $2.5 \times 10^{-1}$ | 12/12 | $3.1 \times 10^{-1}$ |
| $\gamma$ | 1.02 | 1/12 | $2.4 \times 10^{-2}$ | 1/12 | $2.4 \times 10^{-2}$ | 1/12 | $2.4 \times 10^{-2}$ |
| $\gamma$ | 1.05 | 3/12 | $5.5 \times 10^{-2}$ | 3/12 | $5.5 \times 10^{-2}$ | 3/12 | $5.5 \times 10^{-2}$ |

cuits costed by a generic decomposition bound; and sizes ($N \leq 16$, 16-site cylinders) at which classical methods are exact, so the fitted exponents are descriptive. The gap range that can be held fixed across $N$ (0.30–0.55) lies in the easy regime, well above the gaps of longer natural chains.

## CODE AND DATA AVAILABILITY

The `sbpipe` package, the study scripts, all raw results and the sources of this paper are available at https://github.com/aaronseymour7/FilterPipeline.

### Appendix A: Dyadic versus optimized SBC, and cost maps

The dyadic ladder has zero phases, so it passes the ground state with $F(0) = 1$ and its length grows in whole pulses. The optimized design is shorter, at the price of $F(0) < 1$: its success probability is a median 0.88 times $\gamma$ (as low as 0.52). Per shot it uses a median 0.58 times the dyadic evolution time and 0.62 times its Trotter steps, at the same tolerated step. Per success it is a median 0.89× the dyadic cost (range 0.58–1.55) and the cheaper one in 169 of 222 cases. Counting the cheapest of the three filters case by case, the optimized SBC filter wins 51 of 74 cases at $\epsilon = 10^{-2}$ (dyadic the other 23), 57 at $10^{-3}$ (dyadic 16, QETU 1), and 32 at $10^{-4}$, where QETU takes 37 and dyadic 5, all with QETU at $\tau = \pi/2$. The price of the optimized schedule is its design: a nonconvex optimization of about 40 s per case that had to be repeated with other seeds in 21 of 286 points.

At fixed $N$ and $\epsilon = 10^{-2}$ (Fig. 6), the contours of the dyadic cost per success run nearly parallel to the gap axis for $g \gtrsim 0.40$: there the overlap alone sets the cost, from 2–6 × $10^3$ CNOTs at $\gamma = 0.9$ to 1–4 × $10^5$ at $\gamma = 0.03$; below $g \approx 0.40$ the gap starts to matter. The optimized filter is cheaper over most of the map, and its contours tilt toward the gap axis, as its steeper gap exponent implies. At fixed gap and overlap (Fig. 7), the cost per success grows 7–21× (dyadic) from $\epsilon = 10^{-2}$ to $10^{-5}$ and 1.8–4.6× from $N = 8$ to 14, so over this range the target matters more than the size.

### Appendix B: Reproducing the results

The code, raw results and paper sources are at https://github.com/aaronseymour7/FilterPipeline. The controlled study is run by `sbpipe_study/run_gapov.py` and summarized by `derive_gapov_filter.py`; the benchmarks against the MPS alone and the QETU degree analysis by `run_empirical.py`; the QETU query-length study by `run_qetu_tau.py`; the DMRG-only runs by `run_aposteriori.py`, the DMRG error-bar test by `run_margin.py`, and the wrong-input study by `run_extended.py`. `make_figures.py` writes the tables and numerical macros, and `make_figures_measured.py` the measured-only figures and fit table used here. `python -m pytest sbpipe/tests` checks the components, including the bond block of Eq. (3) and the Dirichlet identity (2).

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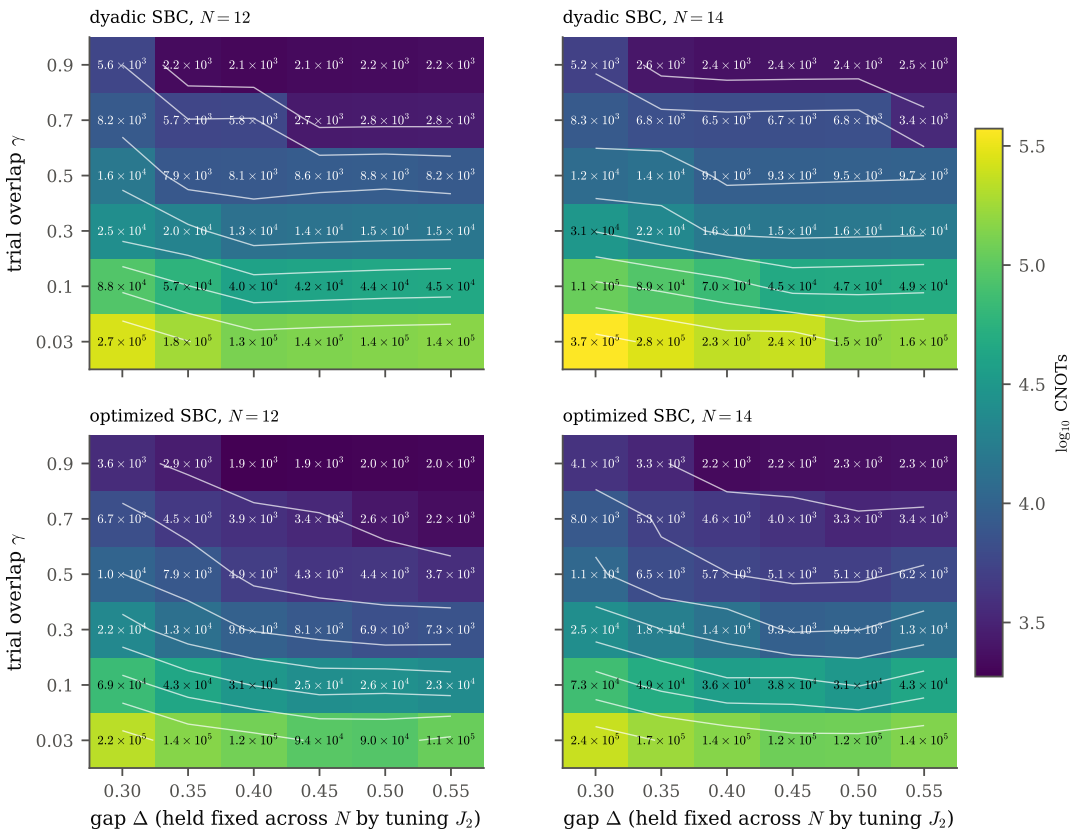



FIG. 6. Filter CNOTs per success over gap and overlap at $\epsilon = 10^{-2}$, measured, for the dyadic (top) and optimized (bottom) SBC filters at $N = 12$ and 14. White contours every factor 1.8. Source: `results/gapov.jsonl`, `results/gapov_grid.jsonl`.


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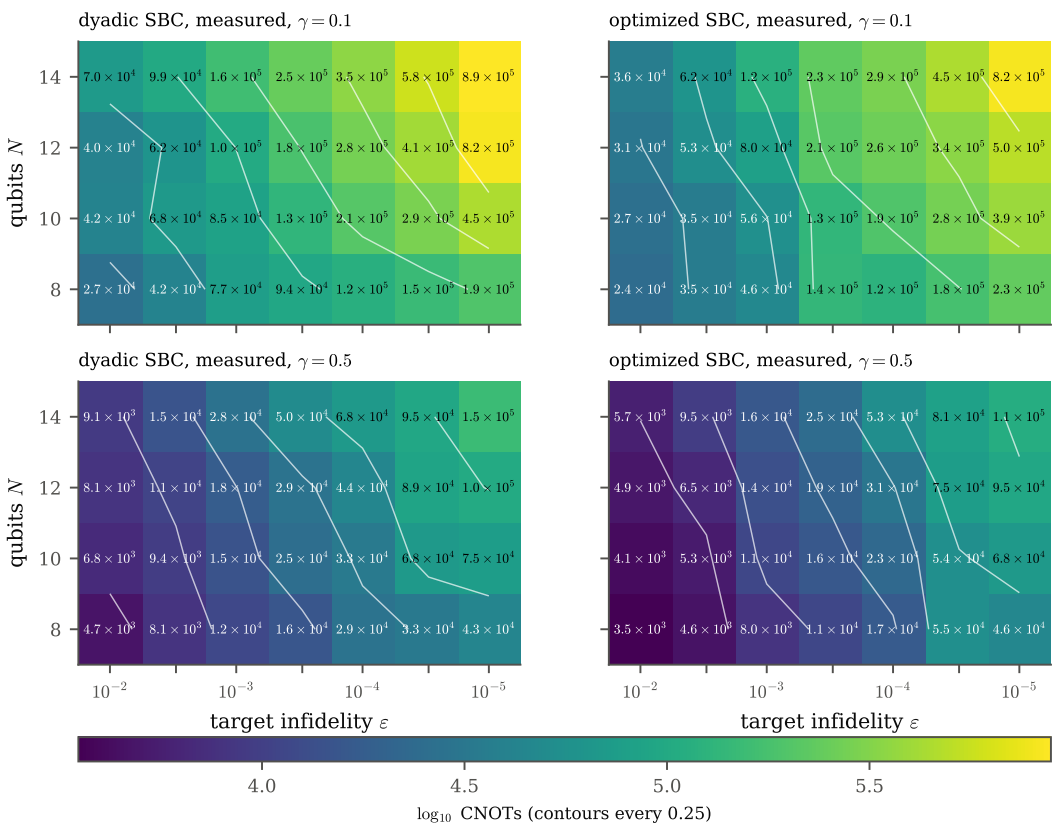



FIG. 7. Filter CNOTs per success over $N$ and $\epsilon$ at a gap held at $g = 0.40$, measured, for the dyadic (left) and optimized (right) SBC filters at $\gamma = 0.1$ (top) and 0.5 (bottom). Source: `results/gapov*.jsonl`.


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